WORKED SOLUTION

STEP II 1994, Question 12

Probability and Statistics • Conditional Probability

STEP II 1994, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q12

The key idea

Once the first throw has fixed a "point" s, every later throw is the same independent experiment with three outcomes: roll s (win), roll 7 (lose), or roll anything else (carry on). So, given the point, the throw on which the game ends is geometric, and the whole question is a matter of splitting into cases by the point and adding up with the law of total probability. The final part is the same calculation with a different conditioning event.

Solution

Let S be the sum of two fair dice. Counting the 36 equally likely outcomes,

$$\mathrm{P}(S = s) = \frac{6 - |s-7|}{36}, \qquad s = 2, 3, \ldots, 12,$$

so the sums $2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ have probabilities $\frac{1}{36} \times (1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1)$.

Winning on the first throw

She wins at once if the first sum is 7 or 11:

$$\mathrm{P}(\text{win on throw 1}) = \frac{6}{36} + \frac{2}{36} = \frac{2}{9}.$$

Winning on the nth throw, given more than one throw

She throws more than once exactly when the first sum is one of the points $4, 5, 6, 8, 9, 10$, which has probability

$$\mathrm{P}(M) = \frac{3+4+5+5+4+3}{36} = \frac{2}{3},$$

where M is the event "she throws more than once".

Fix a point s and write $p_s = \mathrm{P}(S = s)$. Each later throw is independent of the earlier ones and, on it,

$$\mathrm{P}(\text{sum } s) = p_s, \qquad \mathrm{P}(\text{sum } 7) = \tfrac{1}{6}, \qquad \mathrm{P}(\text{neither}) = r_s = 1 - p_s - \tfrac{1}{6}.$$

(Since $s \neq 7$ these two events are disjoint.) She wins on throw $n \geq 2$ exactly when throws $2, \ldots, n-1$ are all "neither" and throw n shows s. So, by independence,

$$\mathrm{P}(\text{win on throw } n \mid \text{point } s) = r_s^{n-2}\, p_s.$$

Points come in pairs with equal probability (4 and 10, 5 and 9, 6 and 8). Write $p_s = k/36$ with $k = 3, 4, 5$; then $r_s = (30-k)/36$, which is $\frac{3}{4}$, $\frac{13}{18}$, $\frac{25}{36}$ respectively. By the law of total probability, for $n \geq 2$,

$$\mathrm{P}(\text{win on throw } n) = \sum_s p_s \cdot r_s^{n-2} p_s = 2\sum_{k=3}^{5} \Big(\frac{k}{36}\Big)^2 \Big(\frac{30-k}{36}\Big)^{n-2}.$$

Winning on throw $n \geq 2$ implies M, so dividing by $\mathrm{P}(M) = \frac{2}{3}$ multiplies each coefficient by $\frac{3}{2}$:

$$2 \cdot \frac{k^2}{1296} \cdot \frac{3}{2} = \frac{k^2}{432} \quad \text{gives} \quad \frac{9}{432} = \frac{1}{48}, \quad \frac{16}{432} = \frac{1}{27}, \quad \frac{25}{432}.$$

Hence

$$\mathrm{P}(\text{win on throw } n \mid M) = \frac{1}{48}\Big(\frac{3}{4}\Big)^{n-2} + \frac{1}{27}\Big(\frac{13}{18}\Big)^{n-2} + \frac{25}{432}\Big(\frac{25}{36}\Big)^{n-2},$$

as required.

> **Ask yourself** Why is it wrong to work with a single "average" point and one geometric distribution?

Because the continuation probability r_s depends on the point. The mixture of three geometric distributions is not itself geometric, which is why the answer has three separate powers rather than one.

Given more than m throws

Let M_m be the event "she throws more than m times", with $m > 1$. This happens exactly when the first throw gives a point s and throws $2, \ldots, m$ are all "neither", so

$$\mathrm{P}(M_m) = \sum_s p_s\, r_s^{m-1} = \frac{2}{36}\sum_{k=3}^{5} k\Big(\frac{30-k}{36}\Big)^{m-1}.$$

If $n \leq m$, winning on throw n is incompatible with M_m, so the conditional probability is 0. If $n > m$, winning on throw n implies M_m, so

$$\mathrm{P}(\text{win on throw } n \mid M_m) = \frac{\mathrm{P}(\text{win on throw } n)}{\mathrm{P}(M_m)} = \frac{9(\frac{3}{4})^{n-2} + 16(\frac{13}{18})^{n-2} + 25(\frac{25}{36})^{n-2}}{36\Big[3(\frac{3}{4})^{m-1} + 4(\frac{13}{18})^{m-1} + 5(\frac{25}{36})^{m-1}\Big]},$$

where the factors $\frac{2}{1296}$ and $\frac{2}{36}$ have been cancelled. Equivalently, multiplying top and bottom by $\frac{1}{432}$,

$$\frac{\frac{1}{48}\left(\frac{3}{4}\right)^{n-2}+\frac{1}{27}\left(\frac{13}{18}\right)^{n-2}+\frac{25}{432}\left(\frac{25}{36}\right)^{n-2}}{\frac{1}{4}\left(\frac{3}{4}\right)^{m-1}+\frac{1}{3}\left(\frac{13}{18}\right)^{m-1}+\frac{5}{12}\left(\frac{25}{36}\right)^{m-1}}, \qquad n > m.$$

Note that the denominator is $\mathrm{P}(M_m \mid M)$, as it should be: the numerator is the earlier answer, conditioned on M, and $M_m \subseteq M$.

Checks

- With $m = 1$ the denominator becomes $\frac{1}{4}+\frac{1}{3}+\frac{5}{12} = 1$, recovering the previous part.
- For $n = 2$ the conditional formula gives $\frac{1}{48}+\frac{1}{27}+\frac{25}{432} = \frac{50}{432} = \frac{25}{216} \approx 0.1157$; directly, $\frac{3}{2}\cdot\frac{2(9+16+25)}{1296}$ gives the same.
- For $m = 2$, $n = 3$ the answer is $\frac{107}{930} \approx 0.1151$. We checked all the formulae against exact step-by-step propagation of the game with fractions, and a simulation of 4×10^5 games agreed to within sampling error.

Common slips

- Forgetting to divide by $\mathrm{P}(M) = \frac{2}{3}$, which gives coefficients $\frac{1}{72}$, $\frac{2}{81}$, $\frac{25}{648}$.
- Using $\mathrm{P}(M_m) = \big(\text{something}\big)^{m-1}$ with a single ratio: the conditioning event is a mixture over points too.
- Counting the "neither" throws as $n-1$ rather than $n-2$: throw 1 sets the point and throw n is the win.

Beyond STEP

The game is a small Markov chain with absorbing states "win" and "lose", and the time to absorption here is a mixture of geometric distributions. Conditioning on survival past m throws shifts the mixture weights towards the points that last longest (here 4 and 10, the least likely to be repeated), a simple example of how a mixture of memoryless distributions is not itself memoryless. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at this level.

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